

# ACHYUTH REDDY MANKU

UNITED STATES | (513) 799-8391 | [achyuthmanku@gmail.com](mailto:achyuthmanku@gmail.com) | [LinkedIn](#)

## PROFESSIONAL SUMMARY

Software engineer with 4+ years building the backend that quietly keeps banks running - mostly Java, Spring Boot, and Node.js services on AWS and Azure. Recently I have been deep in Generative AI work: wiring up OpenAI APIs, building RAG pipelines with LangChain, and turning messy internal documentation into knowledge assistants engineers actually use. I care about clean APIs, fast release cycles, and shipping things that hold up under real production traffic.

## SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, SQL, Go
- **Backend:** Spring Boot, Spring MVC, Spring Security, Hibernate, Node.js, NestJS, REST APIs, GraphQL, Microservices
- **Frontend:** React, Redux, React Hooks, Angular, HTML5, CSS3, Responsive Web Design
- **Generative AI:** OpenAI APIs, LangChain, LangSmith, RAG, Prompt Engineering, Semantic Search, Vector Databases (ChromaDB, Pinecone), TensorFlow.js, Cohere Re-ranking
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, IAM, CloudWatch), Azure, Docker, Kubernetes, Jenkins, Maven, Gradle, Terraform, CI/CD
- **Databases:** PostgreSQL, MySQL, Oracle, MongoDB, Cassandra, DynamoDB, Redis
- **Messaging:** Apache Kafka, RabbitMQ
- **Security:** OAuth 2.0, JWT, RBAC, Spring Security, TLS, Secure SDLC
- **Observability:** Splunk, Grafana, Prometheus, ELK Stack, New Relic
- **Testing & Quality:** JUnit, Mockito, Selenium, JMeter, TDD, SonarQube
- **Tools & Methods:** Git, GitHub, Bitbucket, GitLab, Jira, Confluence, Agile, Scrum, SDLC

## EXPERIENCE

### Citibank

Nov 2024 - Present

#### Software Engineer

- Designed and shipped six cloud-native microservices using Java, Spring Boot, Node.js, and NestJS on AWS and Azure, achieving 99.9% availability to support real-time money transfers
- Built secure REST APIs with OAuth 2.0, JWT, and RBAC for 4 internal banking platforms, which cut cross-team integration time by roughly 30% and saved a lot of back-and-forth in code reviews.
- Rebuilt parts of the internal banking dashboard in React and Angular so it loads cleanly on the devices ops actually use; load times dropped about 25% and the support tickets about 'slow UI' mostly stopped.
- Stitched together a distributed data layer across Cassandra, MongoDB, MySQL, and DynamoDB - picking the right store for each access pattern improved read speeds by ~30%.
- Containerized 10+ services with Docker and Kubernetes and cleaned up the rollout strategy, dropping rollback incidents by 40%.
- Replaced a fragile 2-week release process with Jenkins, Maven, and Gradle pipelines that now ship in 3 days - and we still own zero-downtime deployments.
- Led the build of an internal knowledge assistant using LangChain, RAG, and OpenAI APIs so engineers could ask plain-English questions across policies, runbooks, and troubleshooting docs.
- Tuned the prompt orchestration and context ranking until the answers were actually trustworthy - the team estimates it saved each engineer about 3 hours a week of digging through documents.
- Worked closely with platform and compliance to put guardrails, access controls, and proper audit logging around the AI features so they could live in a regulated environment without raising audit flags.
- Set up monitoring and alerting in Splunk, Grafana, and ELK that actually pages on the right things - caught and resolved 15+ anomalies before any customer noticed.

### Capital One

Dec 2023 - Nov 2024

#### Software Engineer

- Designed Java and Spring Boot backend systems for some of Capital One's high-traffic banking flows; average API response times dropped about 35% after the rework.
- Hunted down latency on REST services running on IBM WebSphere - connection pooling and async patterns took p99 from ~800ms down under 300ms.
- Built UI components in React, Redux, and React Hooks with a focus on responsive design; page load time fell ~20% and engagement metrics increased for the customer-facing app
- Tightened up PostgreSQL with smarter indexing and query tuning; slow-query incidents dropped by 60%.
- Improved Splunk dashboards and alert hygiene so on-call wasn't drowning in noise - MTTR for P1 incidents came down ~25%.
- Collaborated in Agile Scrum sprints, conducting code reviews, applying test-driven development, and pair-debugging with Git and JIRA, which improved code quality and reduced defects

### HSBC

Sep 2020 - Aug 2022

## Java Backend Engineer

- Built Java and Spring Boot REST APIs for account management systems used by 1M+ customers, backed by MySQL and MongoDB.
- Added Redis caching to the read-heavy paths and watched average response times drop ~45% almost overnight.
- Pushed test coverage from 52% to 85% using JUnit, Mockito, and Selenium - production defects fell about 30% as a direct result.
- Tuned SQL queries and indexes in MySQL until average execution time was cut in half (~50%).
- Supported weekly zero-downtime releases with Maven, Jenkins, and Docker, and held a 99.7% availability SLA along the way.

## PROJECTS

---

### Enterprise Document Intelligence with Multi-Modal RAG

- Built a production-grade RAG app on LangChain, GPT-4, and ChromaDB that lets users semantically search 50K+ technical documents - PDFs, Word docs, even code repos - with 89% retrieval accuracy.
- Spent real time on the chunking strategy (semantic splits, context-aware windows) and added hybrid dense + sparse retrieval with Cohere re-ranking; answer relevance improved by ~40%.
- Wired in LangSmith for prompt versioning, latency tracking, and a user-feedback loop so the system gets better over time - average response time settled at 2.3 seconds.

### AI-Powered Financial Assistant with Function Calling

- Built a full-stack Spring Boot + React app that uses OpenAI function calling so users can ask, in plain English, things like 'how much did I spend on food last month?' or 'forecast my budget' and get real, grounded answers.
- Trained a TensorFlow expense classifier hitting 87% accuracy across 15 categories, then layered LLM-generated explanations on top so users understand why a charge was categorized the way it was.
- Tuned MySQL and JWT-based auth to comfortably handle 10K+ monthly transactions with sub-200ms responses.

## EDUCATION

---

### University of Cincinnati

*Master of Science, Information Technology*

**Aug 2022 - May 2023**

*Cincinnati, OH*

### Jawaharlal Nehru Technological University Hyderabad (JNTUH)

*Bachelor of Technology, Computer Science*

**Aug 2017 - May 2021**

*Hyderabad, India*

## CERTIFICATION

---

- **AWS Prompt Engineering: Amazon Web Services**
- **Google AI professional certificate: Google**